House of Quality (HoQ)

Polar M400

GPS WATCH QFD

Quality Engineering

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Technical Report by

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Overview:

This paper represents the Quality Function Deployment (QFD) and Quality Engineering methods for analyzing and improving GPS Watch. It does not take into consideration neither luxury and low-cost products nor non-standard or non-certificated products but instead tries to highlight the range of GPS watches produced for middle economic class consumers.

Also, the following technical report attempts to apprehend more about the House of Quality (HoQ) investigating, Quality Function Deployment (QFD), competitive Q-Benching, Identification of customer requirements (VoC). Classification of product and engineering design requirements [Voice of Engineers (VoE)], Relationship Matrix, Ranking of the technical characteristics (HOWS) importance and accurate analysis of the correlations between the technical features methods involved as well.

INTRODUCTION: FROM A BRIEF HISTORY INTO THE QFD

Global Place System (GPS) invention was one of the much useful discovery in the history of us as human been in our infinite galaxy. GPS watch is a device with integrated GPS receiver that worn as a single block strapped to the wrist, in the manner of a watch. The watch can possess other features and capabilities depending on its intended purpose. GPS watches often practiced for sports and fitness purposes. Many can connect to external sensors by the wireless ANT+ protocol, or a computer by USB to transfer data and configuration. Common sensors used are heart rate monitors and foot pods (running cadence and speed sensor).

SECTION 1 – WHAT CUSTOMER WANTS

The first fundamental step to create the House of Quality (HoQ) is to determine customer requirements (WHAT customer wants) on the other hand the collection of the voices or needs of the customer (VoC). The next bubbles illustrate a list of collected data concerning with GPS-Watch consumers' needs:

- "It should be comfortably wearable."
- "It should react quickly."
- "It should be waterproof/dustproof/shockproof."
- "It should be visible in the dark."
- "It should position accurately."
- "Map should be available."
- "It should have many functions "
- "It should be fashionably wearable."
- "The battery should be durable."
- "Function usage should be friendly."
- "It should be suitable for every situation in life."

To achieve **a clear idea about VoC**, that is helpful to divide and categorize them:

Structure & Appearance (Comfortable and Fashionable)
Durability (Good Battery / Rechargeable, Resistance Against External Conditions)
Usage (Accuracy, User friendly / Compatibility, Responsiveness / Reaction Time)
Function (Light, Additional Features, Map Inclusion / Map Availability)

In addition, another important thing to do is Identification of the most important basic requirements. Table 1.1 in following illustrates an example of how they could stir from the level 1 (one) of customers' statements (VoC) into level 3 with a more detailed definition of the first key clients' requirements. Either, there are several techniques such as customer requirements tree, affinity diagrams, hierarchical cluster analysis and KJ method that would be manipulating purely basic customer requirements typically, can be grouped into four macro-categories.

100 It should be suitable for every situation in life	110 Daily situations	111 Comfortably wearable
		112 Fashionable
	120 Extreme situations	121 Usable at a wide temperature range
		122 Resistance to water/dust/shock
200 It should have a lot of functions	210 Basic usage	211 Accurately locates
		212 User Friendly
		213 Quickly responds
	220 Affiliated functions	221 Light watch
		222 Additional features(health/sports)
		223 Map inclusion

Table 1.1: VoC transition from first level to third

SECTION 2 – HOW MEET CUSTOMER NEEDS

The second fundamental step to create QFD or someone wants to benefits from a Fuzzy-QFD approach is the determination of the technical characteristics (HOW to meet customer needs). Customer requirements expressed in subjective terms must translate into objective technical characteristics or technical features.

Structure & Appearance

- Comfortable → Case / Wristband material; Manufacturing tolerances; Mass; Display size
- Fashionable → Case / Wristband material; Manufacturing tolerances; Display / Hardness of the Display; Number of colors; size; Quality / Quantity of the sensors.

It is possible to observe the following list of collected data with respect to GPS watch technical characteristics and (Voice of Engineers):

- Case / Wristband Material
- Inner Material
- Manufacturing Tolerances
- Battery Capacity
- Strength Of The Antenna
- Display / Hardness Of The Display
- Time For Software Development
- Strength Of The CPU
- Energy Consumption Of The Watch
- Mass
- Number Of Colors
- Display Size
- Size Of The Memory (Internal)
- Quantity / Quality Of The Sensore

SECTION 3 – RELATIONSHIP MATRIX

The third fundamental step is the creation of relationship matrix. It must establish how the technical decisions affect the satisfaction of each customer's need. The relationships intensities are 9 (●) for strong coordination, 3 (○) for intermediate and 1 (△) for weak coordination.

	Customer Requirements / Technical Characteristics	Degree of importance	Relative importance	Case / Wristband material	Inner material	Manufacturing tolerances	Battery capacity	Strength of the antenna	Display / Hardness of the display	Time for software development	Strength of the CPU	Energy consumption	Mass	Number of colours	Display size	Memory size	Quality / Quantity of the sensors
				1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	Comfortability	3,4	9,5%	●		△							●		○		
2	Accuracy	4,4	12,3%		△	△		●		●	●	○				△	●
3	Good battery / Rechargable	3,8	10,6%				●	○			△	●			○		△
4	Light watch	2,9	8,1%	●	○		○	△					●		○	△	△
5	Fashionable	2,8	7,8%	●		△			△					●	○		△
6	Resistance against external conditions	3,4	9,5%	●	△	○			●						△		○
7	Userfriendlyness / Compatibility	4	11,2%							●					●	△	●
8	Additional features (health/sport)	3	8,4%							●	○	○			△	○	●
9	Responsiveness / Reaction time	4	11,2%		○					●	●	○					○
10	Map inclusion / Map availability	4	11,2%							●						○	

Technical Importance	3,15	0,80	0,58	1,20	2,52	0,94	4,89	2,48	1,92	1,59	0,71	2,27	0,90	3,76
Technical relative imp.	11,4%	2,9%	2,1%	4,3%	9,1%	3,4%	17,7%	8,9%	6,9%	5,7%	2,5%	8,2%	3,3%	13,6%
Absolute weight	3,29	0,82	0,68	1,03	2,46	1,23	4,93	2,53	1,81	1,52	0,60	2,03	0,97	3,92
Relative weight	11,8%	2,9%	2,4%	3,7%	8,9%	4,4%	17,7%	9,1%	6,5%	5,5%	2,2%	7,3%	3,5%	14,1%
Unit of Measurement		L	%	mAh	dBm	HRC	Month	Ghz	µAη/η	g	Nr	Pixels	MB	
Polar M400	-	-	0,2	190	21	-	5	0,8	24	56.6	2	128x128	500	-
Garmin Forerunner 15	-	-	0,23	100	27	-	8	1,4	11	36	5	55 x 32	200	-
TomTom Multi-Sport	-	-	0,8	120	27	-	10	2,4	12	62.9	1	144x168	500	-
Targets (new model)	-	-	**0.2**	**120**	**27**	-	**10**	**2**		**36**	**5**	**144x168**	**500**	-

Table 3.1: GPS Watch House of Quality

SECTION 4 – PRIORITIZATION OF CUSTOMERS' REQUIREMENTS

The fourth fundamental step is the prioritization of customers' perceived quality of each requirement[1]. Because many of the quality characteristics evaluated by the client might be measured quantitatively by the service company, the procedures ensuring service quality must anticipate the definition of a system of indicators able to verify them, by monitoring the numerical values inherent in the selected reference measurements[2]. The "degree of importance" or evaluation of the characters rank from one (1) - not considered a strength, two (2) - a possible strength, down to five (5) - a significant strength. In this section, the analysis will cover and include the Benchmarking by perceived work condition quality and "Target values of expectations." Recognizing, studying, identifying and analyzing the competitors in this step of Quality Function Deployment (QFD) creation is strongly recommended. Identifying or analyzing competitors [Q-Benching], the devotion of time and attention to acquiring knowledge on them (and their products they already produce) in an abstract method especially by analyzing the consumer's psyche is imperative.

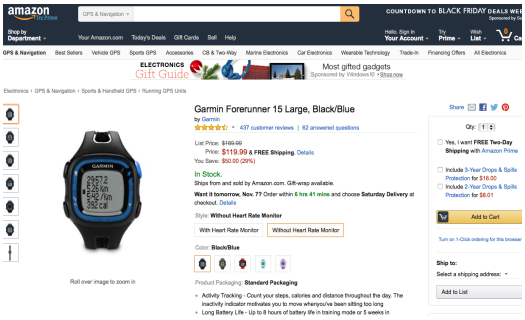
[1] Chapter 7 of "Advanced quality function deployment" by Fiorenzo Franceschini; 2002

[2] ISO 9004/2, 1991

In the following section (SECTION 4.1, 4.2), some additional detailed information about GARMIN and TOMTOM are illustrated. These two famous companies are blood rival competitors in the GPS Watches Market, and both of them are thoroughly well-known or popular to produce GPS Watch under the $250 cost.

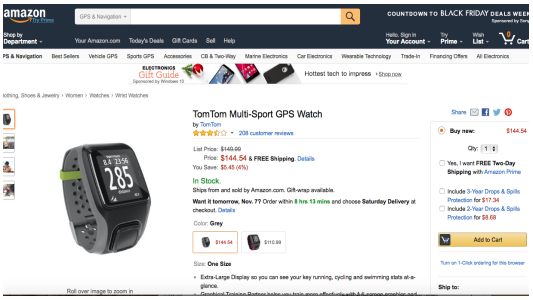
SECTION 4.1 – GARMIN FORERUNNER 15

Garmin is one of the most prominent GPS producers in the world. It manufactures a huge range of different use GPS's from airplane navigating components to the climbing GPS. However, the Forerunner 15 faced an overwhelming demand from the GPS consumers' part. Runners could run with all their hearts and let FORERUNNER 15 captures essential stats from their run like time, distance or pace when paired with a heart rate strap system.



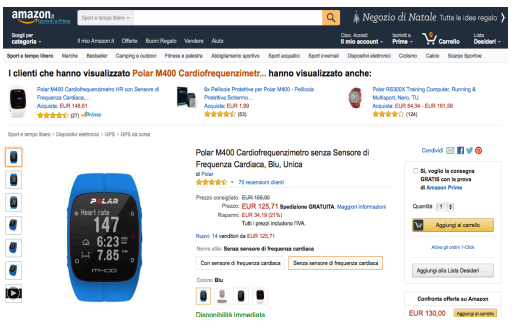
SECTION 4.2 – TOMTOM MULTI-SPORT

TomTom Company recognized widely because of its vehicle navigation systems. After all these success TomTom company decided to run and compete even in GPS Watch market as a new business but not so far as a new segment. The TOMTOM MULTI-SPORT is one of the best user-friendly or simple GPS watch product has ever manufactured by TomTom Company.



SECTION 4.3 – SPECIFICATION OF OUR PRODUCT (POLAR M400)

The Polar company has known by many sportive people as a prolific white bear in an extensive world of sports. Particularly in some kind of sports such as sailing, sea rafting, snowboarding, running, climbing, skying and skiing. However, POLAR M400 is the GPS running watch that is developed for runners who want a comfortable and durable running watch with sporty design and advanced GPS and training features.



SECTION 5 – TECHNICAL IMPORTANCE RANKING

The fifth step consists of analyzing and calculating the Technical Importance of the Technical Characteristics in the above-mentioned relationship matrix. Additionally, the key calculations made to compute Technical Importance Percentage, Technical Relative Importance, Absolute Weight, and Relative Weight.

SECTION 5.1 – LYMAN'S NORMALIZATION

Lyman's normalization of data and the normalization methods[3] are used to identify the most suitable normalization procedures to apply to the problem and to take into account their properties concerning the measurement units in which the indicators express their robustness against possible outliers. The main feature of Lyman's normalization is that every single value used to compute the final percentages, which are "relativized" to the sum of each row considering the numbers assigned using the Independent Scoring Method (ISM).

	Technical relative imp.				
	No Lyman		Lyman		
Time for software development	17,65%	1	20,15%	1	Time for software development
Quality / Quantity of the sensors	13,58%	2	12,57%	2	Case / Wristband material
Case / Wristband material	11,38%	3	11,98%	3	Quality / Quantity of the sensors
Strength of the antenna	9,09%	4	8,59%	4	Display size
Strength of the CPU	8,94%	5	6,94%	5	Strength of the antenna
Display size	8,20%	6	6,75%	6	Strength of the CPU
Energy consumption	6,92%	7	6,40%	7	Energy consumption
Mass	5,73%	8	6,33%	8	Mass
Battery capacity	4,34%	9	4,67%	9	Memory size
Display / Hardness of the display	3,38%	10	4,50%	10	Battery capacity
Memory size	3,27%	11	3,62%	11	Display / Hardness of the display
Inner material	2,88%	12	2,94%	12	Number of colours
Number of colours	2,55%	13	2,41%	13	Inner material
Manufacturing tolerances	2,10%	14	2,15%	14	Manufacturing tolerances

Table 5.1: No Lyman's versus Lyman's normalization method

This is the unique difference in the computing procedure concerning the classical method. Meanwhile, Table 5.1 illustrates some technical characteristics those highlighted because they change mostly after Lyman's Normalization. The same approach applied to Table 5.2 that keeps in consideration the relative weight.

	Relative weight				
	No Lyman		Lyman		
Time for software development	17,74%	1	20,35%	1	Time for software development
Case / Wristband material	11,82%	2	12,99%	2	Case / Wristband material
Quality / Quantity of the sensors	14,10%	3	12,48%	3	Quality / Quantity of the sensors
Display size	7,29%	4	7,66%	4	Display size
Strength of the antenna	8,86%	5	6,89%	5	Strength of the CPU
Strength of the CPU	9,09%	5	6,69%	6	Strength of the antenna
Energy consumption	6,50%	7	6,01%	7	Mass
Mass	5,47%	8	5,95%	7	Energy consumption
Display / Hardness of the display	4,43%	9	4,92%	9	Memory size
Battery capacity	3,70%	10	4,76%	10	Display / Hardness of the display
Memory size	3,49%	11	3,83%	11	Battery capacity
Inner material	2,94%	12	2,50%	12	Number of colours
Manufacturing tolerances	2,43%	13	2,49%	12	Inner material
Number of colours	2,16%	14	2,48%	12	Manufacturing tolerances

Table 5.2: the differences after applying the normalization

[3] The "Handbook Composite Indicators" OECD, 2008 ; Page 20, 27, 30 and 85 (STEP 5.)

SECTION 5.2 – Q-Bench Applying

In the following section, examples of our Q-Bench alterations will be displayed. Battery strength, energy consumption, weights and display resolution will be the chosen parameters. The selected upper and lower limits derived from good and bad products of direct competitors.

Q-Bench Parameters	Battery strength [mAh]	Energy consumption [mA]	Weight [g]	Display resolution (in pixel)
Upper limit	200	10	36	24292
Lower limit	80	24	62,9	1000

Competitors Parameters	Battery strength [mAh]	Energy consumption [mA]	Weight [g]	Display resolution (in pixel)
Garmin Forerunner 15	100	12.5	36	1760
Tom Tom Multi-Sport	120	12	62.9	24192

Relative calculations regarding Q-Bench applying method are the sequence:

$$W1 = (10.6 * 9 + 8.1 * 3)/ (10.6 * 9 + 8.1 * 3)+(12.3 * 3 + 10.6 * 9 + 8.4 * 3 + 11.2 * 3) + (9.5 * 9 + 8.1 * 9) + (9.5 * 3 + 10.6 * 3 + 8.2 * 3 + 7.8 * 3 + 9.5 * 1 + 11.2 * 9 + 8.4 * 1)$$

$$W2 = (12.3 * 3 + 10.6 * 9 + 8.4 * 3 + 11.2 * 3)/ (10.6 * 9 + 8.1 * 3)+(12.3 * 3 + 10.6 * 9 + 8.4 * 3 + 11.2 * 3) + (9.5 * 9 + 8.1 * 9) + (9.5 * 3 + 10.6 * 3 + 8.2 * 3 + 7.8 * 3 + 9.5 * 1 + 11.2 * 9 + 8.4 * 1)$$

$$W3 = (9.5 * 9 + 8.1 * 9)/ (10.6 * 9 + 8.1 * 3)+(12.3 * 3 + 10.6 * 9 + 8.4 * 3 + 11.2 * 3) + (9.5 * 9 + 8.1 * 9) + (9.5 * 3 + 10.6 * 3 + 8.2 * 3 + 7.8 * 3 + 9.5 * 1 + 11.2 * 9 + 8.4 * 1)$$

$$W4 = (9.5 * 3 + 10.6 * 3 + 8.2 * 3 + 7.8 * 3 + 9.5 * 1 + 11.2 * 9 + 8.4 * 1)/ (10.6 * 9 + 8.1 * 3)+(12.3 * 3 + 10.6 * 9 + 8.4 * 3 + 11.2 * 3) + (9.5 * 9 + 8.1 * 9) + (9.5 * 3 + 10.6 * 3 + 8.2 * 3 + 7.8 * 3 + 9.5 * 1 + 11.2 * 9 + 8.4 * 1)$$

W1 = 0.172 W3 = 0.228

W2 = 0.274 W4 = 0.3

In the first profile, we are choosing [200, 24, 65, 1500] and results are following:

	J+	J-	J=	(w+)+(W=)/W- (=k)	W+/W-	oAo' (k > 0.66?)
(ax,ay)	4	1,2,3		0.483	N	N
(ax,a)	3,4	1	2	4.81	Y	Y
(ay,ax)	1,2,3	4		2.067	Y	Y
(ay,a)	2,4	1	3	4.81	Y	Y
(a,ax)	1	3,4	2	0.805	N	Y
(a,ay)	1	2,4	3	0.666	N	Y

Through the cases where both criteria are satisfied with "Yes", we can see that:

ay > ax ay > ax

To beat our competitor we need to raise the next important technical characteristic. For the second alteration in addition to the battery capacity we also improved the energy consumption choosing [200, 10, 65, 1500]:

	J+	J-	J=	(w+)+(W=)/W- (=k)	W+/W-	oAo' (k> 0.66?)
(ax,ay)	4	1,2,3		0.483	N	N
(ax,a)	3,4	1,2		1.24	Y	Y
(ay,ax)	1,2,3	4		2.067	Y	Y
(ay,a)	4	1,2	3	1.24	N	Y
(a,ax)	1,2	3,4		0.805	N	Y
(a,ay)	1,2	4	3	2.067	Y	Y

In this case we receive:

ax > a a > ay ay > ax

This is an incomparable or sovereign state. Because we are still weaker than "ax" and we continue by raising the next parameter. Our next product state defined as the following: [200, 10, 36, 1500]

	J+	J-	J=	(w+)+(W=)/W- (=k)	W+/W-	oAo' (k> 0.66?)
(ax,ay)	4	1,2,3		0.483	N	N
(ax,a)	4	1,2	3	1.24	N	Y
(ay,ax)	1,2,3	4		2.067	Y	Y
(ay,a)	4	1,2,3		0.483	N	N
(a,ax)	1,2	4	3	2.067	Y	Y
(a,ay)	1,2,3	4		2.067	Y	Y

In our third iteration or redundancy, we find out a new result in the product ranking:

ay > ax a > ay a > ax

Put together these results means that a > ay > ax. Additionally, it means that our product (Polar M400 GPS Watch) might strictly outrank both competitors.

SECTION 6 – CORRELATION MATRIX

The sixth step, which is not the essential step, is to create "QFD correlation matrix" analysis and complete the "roof" of the House of Quality (HoQ). For instance, "Case or Wristband Material" and "Display or Hardness Of The Display" as two Technical Characteristics have a positive correlation with another Character such as the "Mass". The sign (+) is a symbol of this relationship between these Voices of Engineers (VoE). Vice versa, there is a negative correlation between the "Mass" and "Battery Capacity", "Strength of Antenna" and "Battery Consumption" those illustrated and represented by the sign (-) which means negative correlation.

SECTION 7 – SUMMARY

The relations or not relationships between any single elements of referred project [QFD for GPS Watch], might be one of the most significant results of QFD analyzing method. Another valuable and acquired result of QFD for "GPS Watch" depicted in recognizing strength and weakness of competitors that might lead manufacturers or engineers to understand the best sustainable further decisions.

Technical Importance	3,15	0,80	0,58	1,20	2,52	0,94	4,89	2,48	1,92	1,59	0,71	2,27	0,90	3,76
Technical relative imp.	11,4%	2,9%	2,1%	4,3%	9,1%	3,4%	17,7%	8,9%	6,9%	5,7%	2,5%	8,2%	3,3%	13,6%
Absolute weight	3,29	0,82	0,68	1,03	2,46	1,23	4,93	2,53	1,81	1,52	0,60	2,03	0,97	3,92
Relative weight	11,8%	2,9%	2,4%	3,7%	8,9%	4,4%	17,7%	9,1%	6,5%	5,5%	2,2%	7,3%	3,5%	14,1%

The three most important values to improve are the Efforts for Software Development with 17,7%, the Case / Wristband Material with 11,8% and the Quality and Quantity of "GPS Watch" sensors with 14,1% Relative Importance rating.

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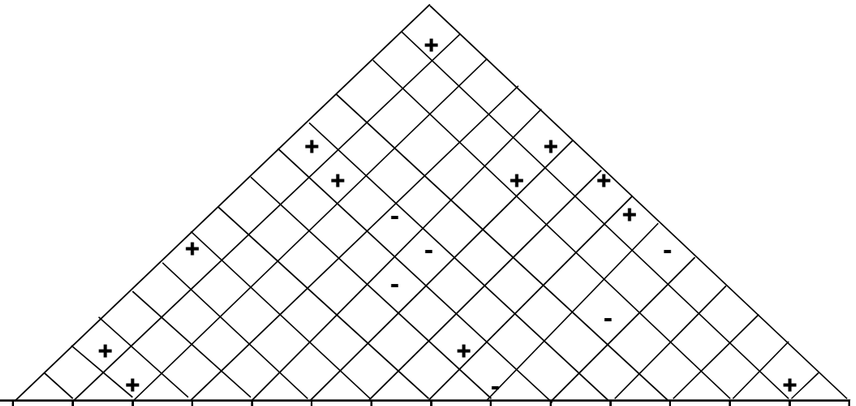
Relationship intensity
● = 9
○ = 3
△= 1



	Customer Requirements / Technical Characteristics	Degree of importance	Relative importance	1 Case / Wristband material	2 Inner material	3 Manufacturing tolerances	4 Battery capacity	5 Strength of the antenna	6 Display / Hardness of the display	7 Time for software development	8 Strength of the CPU	9 Energy consumption	10 Mass	11 Number of colours	12 Display size	13 Memory size	14 Quality / Quantity of the sensors	Polar M400	Garmin Forerunner 15	TomTom Multi-Sport	Targets (new model)	Improvement ratio	Strength	Absolute weight	Relative weight
1	Comfortability	3,4	9,5%	●		△							●		○			3	5	4	4	1,33	1	4,5	8,6%
2	Accuracy	4,4	12,3%		△	△		●		●	●	○				△	●	3	5	4	4	1,33	1,2	7,0	13,4%
3	Good battery / Rechargable	3,8	10,6%				●	○			△	●			○		△	4	5	3	4	1,00	1,2	4,6	8,7%
4	Light watch	2,9	8,1%	●	○		○	△					●		○	△	△	4	5	4	5	1,25	1,2	4,4	8,3%
5	Fashionable	2,8	7,8%	●		△			△					●	○		△	4	5	3	5	1,25	1	3,5	6,7%
6	Resistance against external conditions	3,4	9,5%	●	△	○			●						△		○	3	4	4	4	1,33	1,5	6,8	12,9%
7	Userfriendlyness / Compatibility	4	11,2%							●					●	△	●	4	4	5	4	1,00	1,2	4,8	9,1%
8	Additional features (health/sport)	3	8,4%							●	○	○			△	○	●	4	4	5	5	1,25	1,5	5,6	10,7%
9	Responsiveness / Reaction time	4	11,2%		○					●	●	○					○	3	4	3	4	1,33	1	5,3	10,2%
10	Map inclusion / Map availability	4	11,2%							●						○		4	4	5	5	1,25	1,2	6,0	11,4%

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Technical Importance	3,15	0,80	0,58	1,20	2,52	0,94	4,89	2,48	1,92	1,59	0,71	2,27	0,90	3,76
Technical relative imp.	11,4%	2,9%	2,1%	4,3%	9,1%	3,4%	17,7%	8,9%	6,9%	5,7%	2,5%	8,2%	3,3%	13,6%
Absolute weight	3,29	0,82	0,68	1,03	2,46	1,23	4,93	2,53	1,81	1,52	0,60	2,03	0,97	3,92
Relative weight	11,8%	2,9%	2,4%	3,7%	8,9%	4,4%	17,7%	9,1%	6,5%	5,5%	2,2%	7,3%	3,5%	14,1%
Unit of Measurement		L	%	mAh	dBm	HRC	Month	Ghz	μΛη/η	g	Nr	Pixels	MB	
Polar M400	-	-	0,2	190	21	-	5	0,8	24	56.6	2	128x128	500	-
Garmin Forerunner 15	-	-	0,23	100	27	-	8	1,4	11	36	5	55 x 32	200	-
TomTom Multi-Sport	-	-	0,8	120	27	-	10	2,4	12	62.9	1	144x168	500	-
Targets (new model)	**-**	**-**	**0.2**	**120**	**27**	**-**	**10**	**2**		**36**	**5**	**144x168**	**500**	**-**

COMPARISON		1	2	3	4	5	6	7	8	9	10	11	12	13	14
		ranking technical relative imp. (Normal and Lyman)													
		2	13	14	10	5	11	1	6	7	8	12	4	9	3
		3	12	14	9	4	10	1	5	7	8	13	6	11	2
Technical relative imp.	no Lyman	11,4%	2,9%	2,1%	4,3%	9,1%	3,4%	17,7%	8,9%	6,9%	5,7%	2,5%	8,2%	3,3%	13,6%
	Lyman	12,6%	2,4%	2,2%	4,5%	6,9%	3,6%	20,1%	6,8%	6,4%	6,3%	2,9%	8,6%	4,7%	12,0%
Relative weight	no Lyman	11,8%	2,9%	2,4%	3,7%	8,9%	4,4%	17,7%	9,1%	6,5%	5,5%	2,2%	7,3%	3,5%	14,1%
	Lyman	13,0%	2,5%	2,5%	3,8%	6,7%	4,8%	20,4%	6,9%	6,0%	6,0%	2,5%	7,7%	4,9%	12,5%
		2	12	12	11	6	10	1	5	7	7	12	4	9	3
		2	12	13	10	5	9	1	5	7	8	14	4	11	3
		ranking relative weights (Normal and Lyman)													